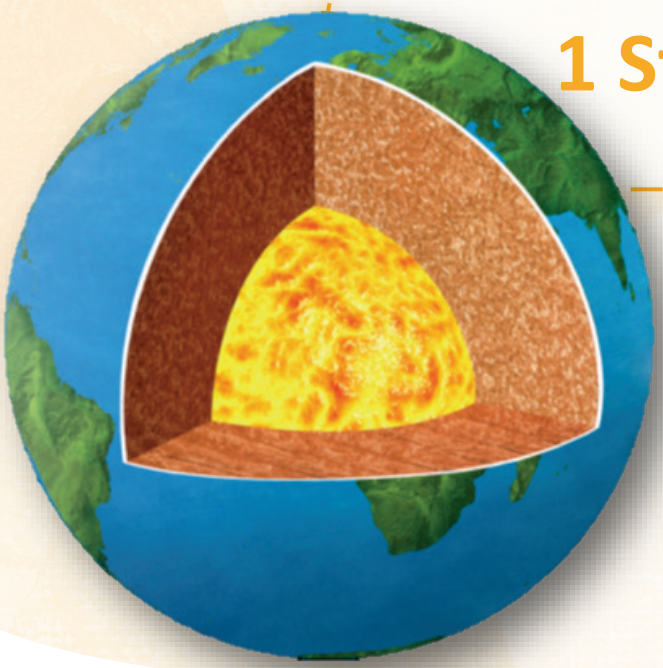
New Science

Revise this chapter together

Unit 1 About the Earth

1 Structure of the Earth

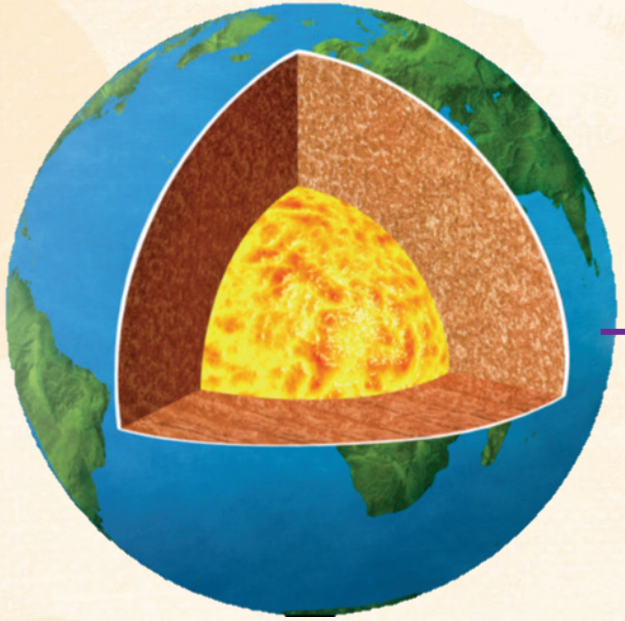




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1. The outermost layers of the Earth is the crust

What is the **structure** of the Earth?



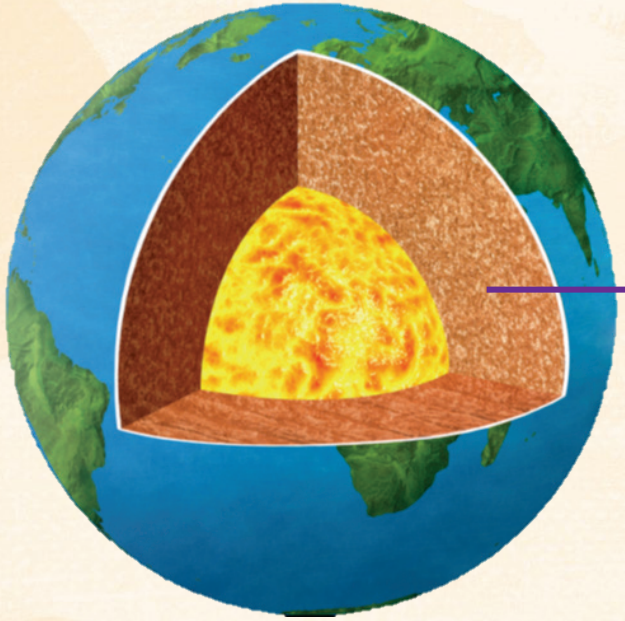
The outermost layer is the crust.

Characteristics:

- mainly made of rocks
- it is divided into many plates

2. Below the crust is the mantle. It contains hot flowing magma.

What is the **structure** of the Earth?



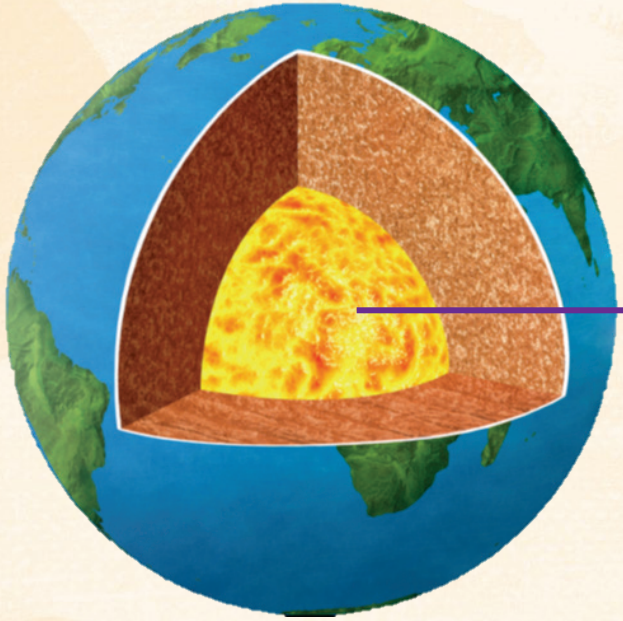
Below the crust is the mantle.

Characteristics:

- it contains hot flowing magma

3. The centre of the Earth is the extremely hot core.

What is the **structure** of the Earth?

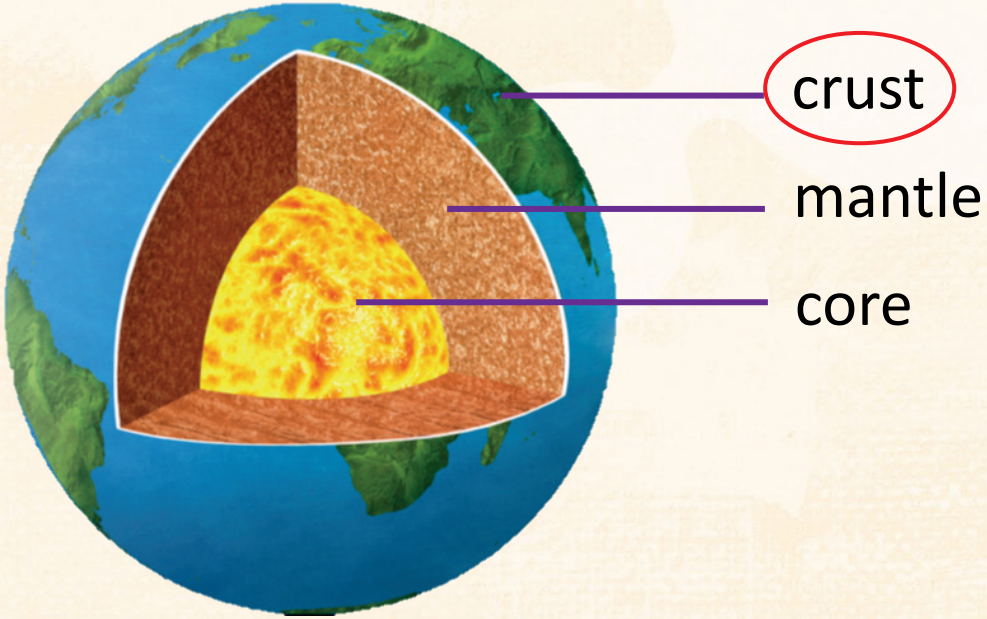


The centre of the Earth is the core.

Characteristics:

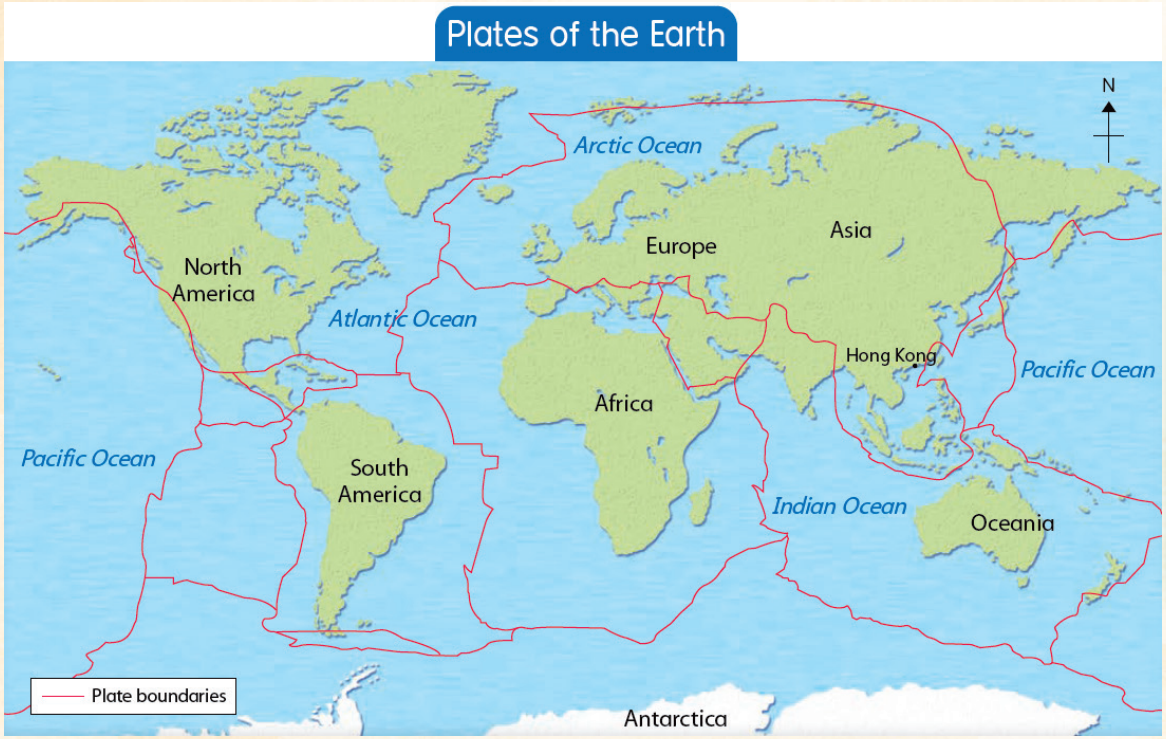
- it is mainly made up of metals
- the temperature is extremely high

In which layer of the Earth do we **live**?



4. The crust can be divided into many plates. Plate movement and collision may cause earthquakes or volcanic eruptions.

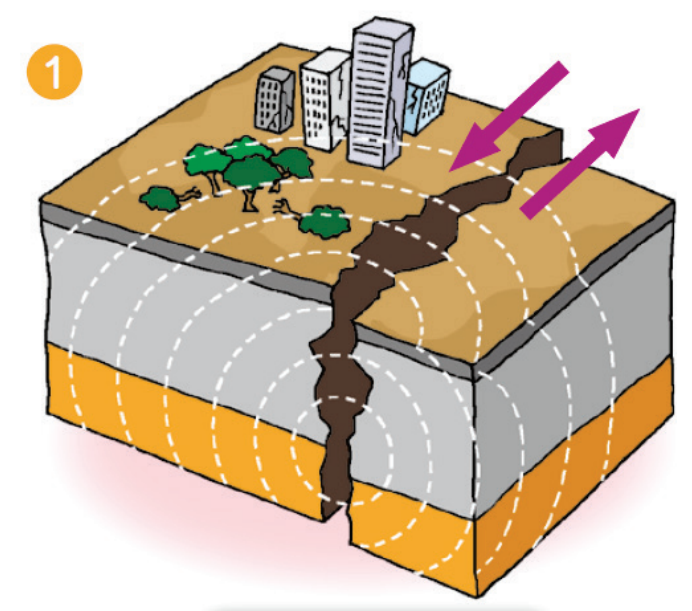
What are the characteristics of the Earth's crust? Is it a **complete** piece?



- The crust is divided into plates of different sizes.
- It is / is not a complete piece.

4. The crust can be divided into many plates. Plate movement and collision may cause earthquakes or volcanic eruptions.

What **natural phenomena** may plate movement and collision cause?

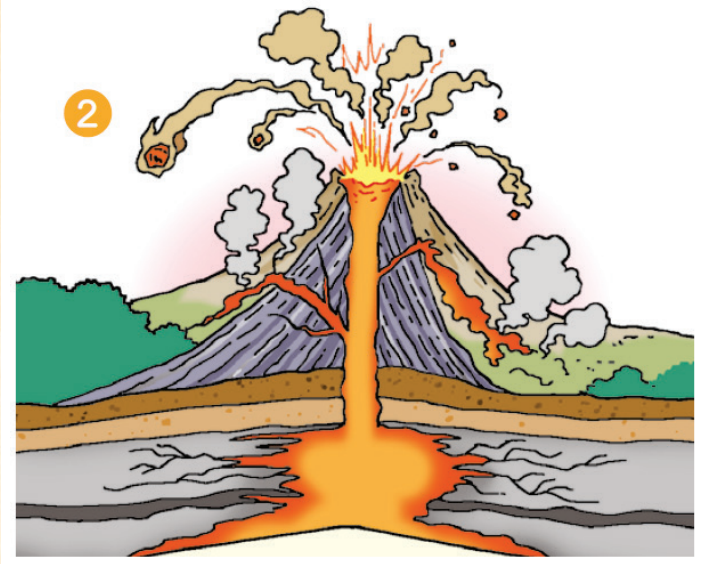


the ground _shakes_ violently

→ causes _earthquakes_

4. The crust can be divided into many plates. Plate movement and collision may cause earthquakes or volcanic eruptions.

What **natural phenomena** may plate movement and collision cause?

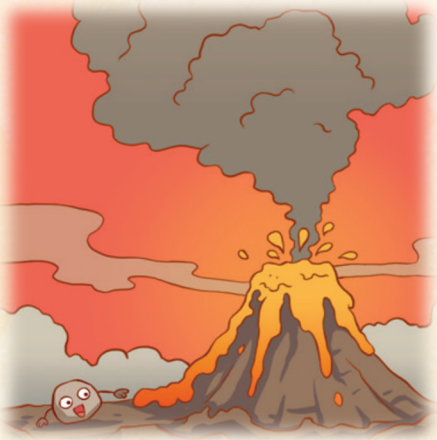


magma gushes out of a volcano

→ causes volcanic eruptions

5. The crust is mainly made up of rocks, which can be classified into three types according to the formation processes.

What is the Earth's crust **made up of**?





volcanic eruptions cause the magma to cool and harden

→ form _rocks_

some rocks broke after being exposed to wind and rain for many years

→ form _soil_

5. The crust is mainly made up of rocks, which can be classified into three types according to the formation processes.

What **types** of rocks can be classified into? What are some examples?

types of rocks

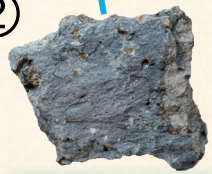
A. igneous rocks

B. sedimentary rocks

C. metamorphic rocks

① 

granite

② 

basalt

③ 

sandstone

④ 

shale

⑤ 

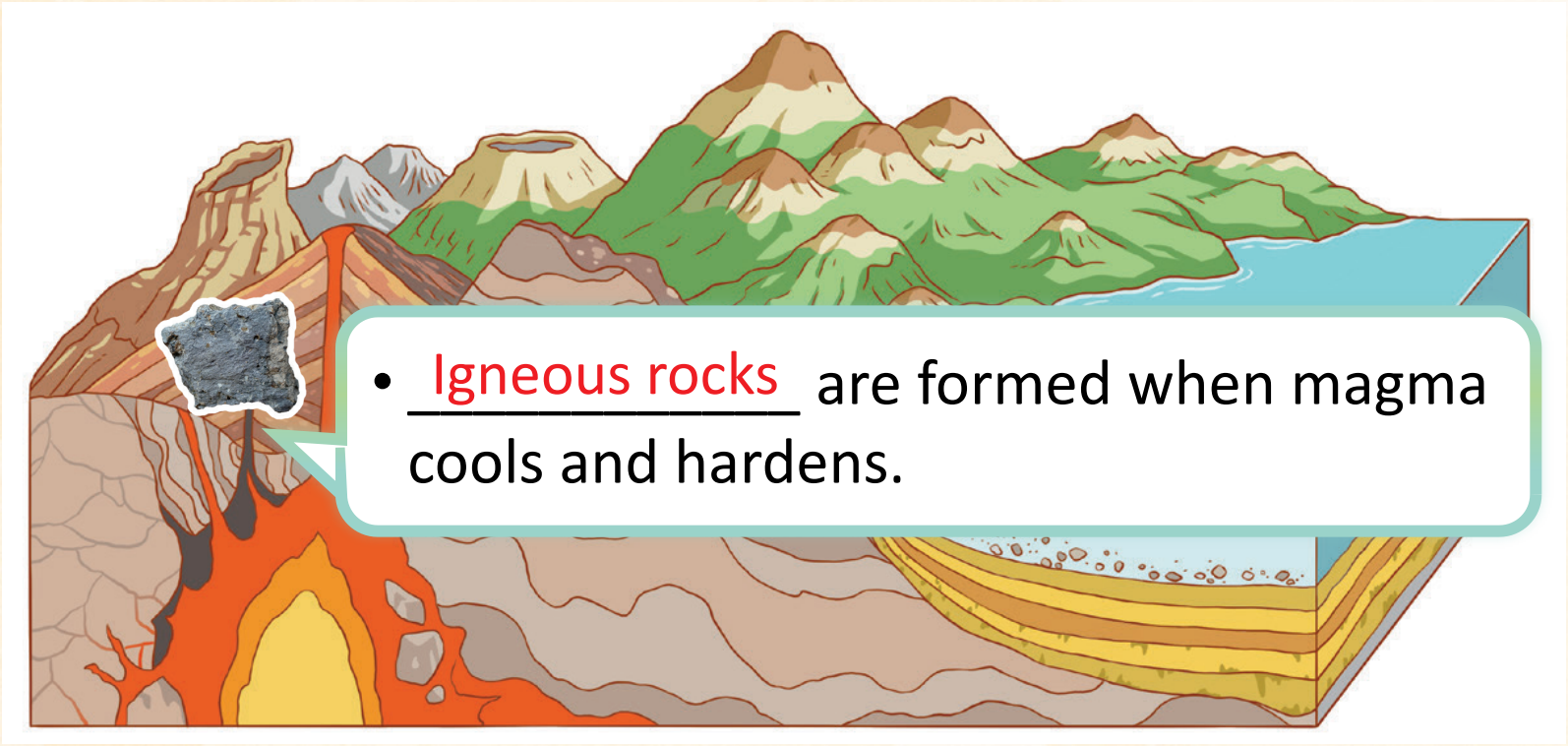
marble

⑥ 

gneiss

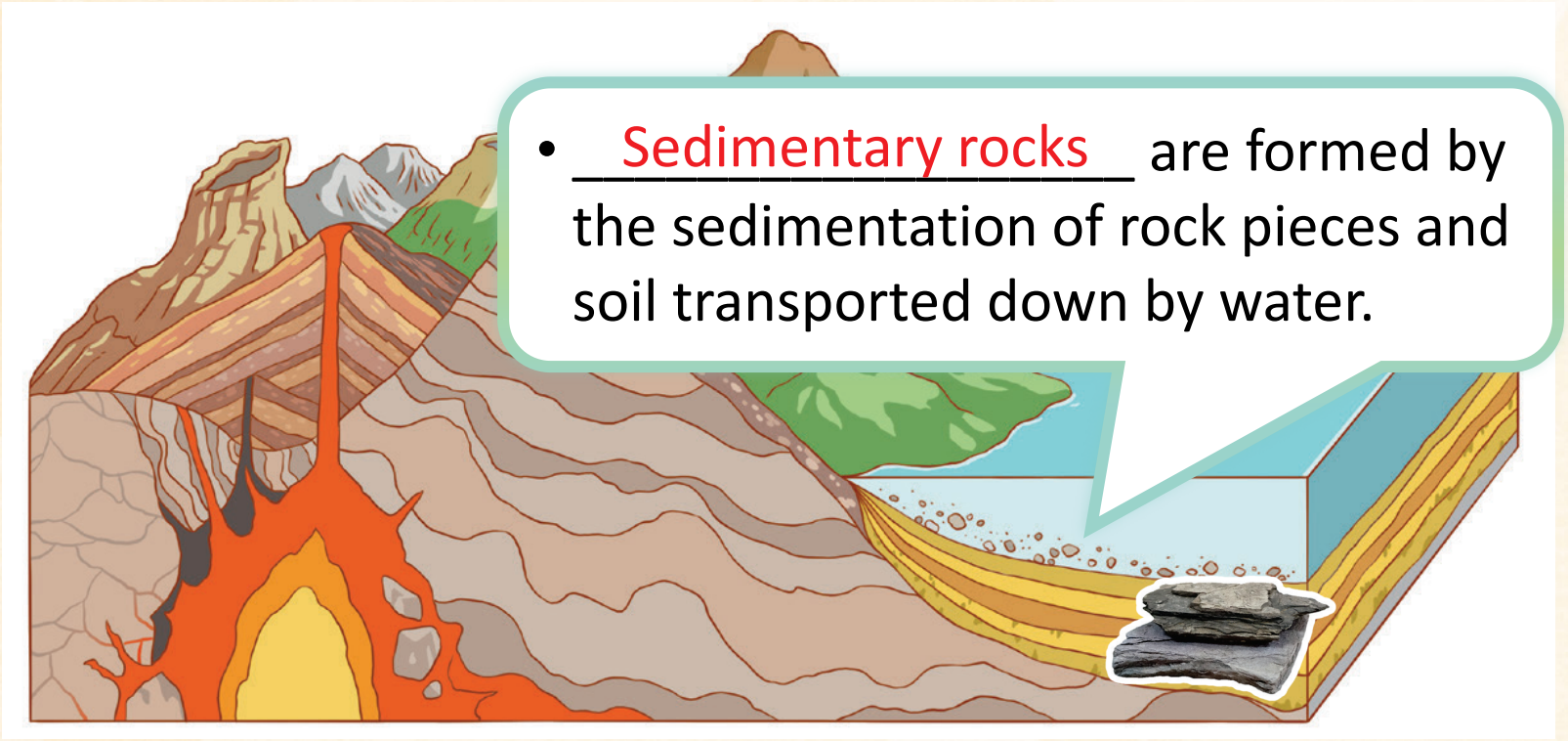
6. Igneous rocks are formed when magma cools and hardens.

How are different types of rocks **formed**?



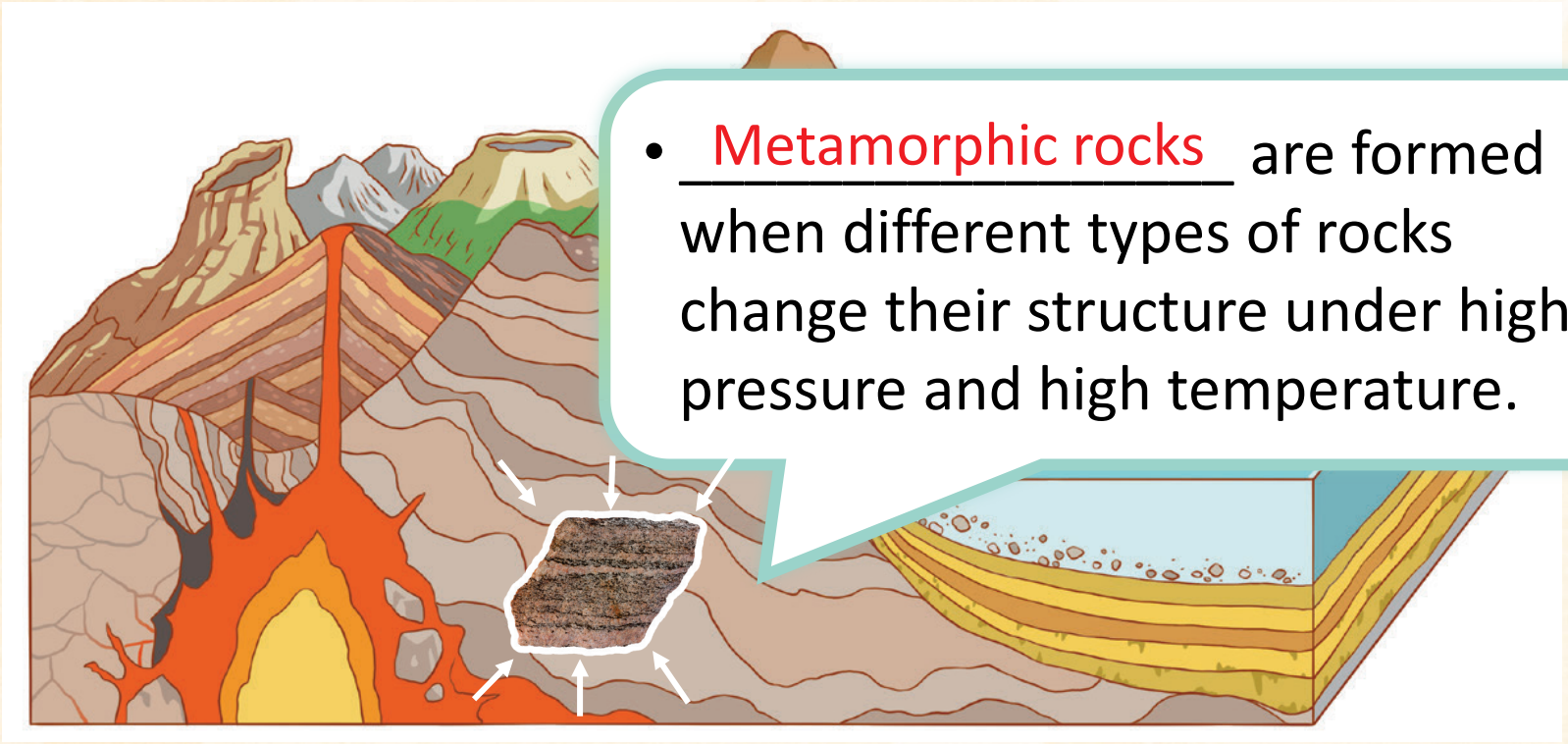
7. Sedimentary rocks are formed by the sedimentation of rock pieces and soil transported down by water.

How are different types of rocks **formed**?



8. Metamorphic rocks are formed when different types of rocks change their structure under high pressure and high temperature.

How are different types of rocks **formed**?



9. Different types of rocks have different colours, grain sizes and hardness. They have different uses in our daily lives.

What are the **characteristics** of different types of rocks?


granite


sandstone


marble

Use tools to compare the **colours**, **grain sizes** and **hardness** of different types of rocks.

- colours: same/ different
- the largest grains: granite
- the hardest: granite/ sandstone

9. Different types of rocks have different colours, grain sizes and hardness. They have different uses in our daily lives.

Based on the characteristics of rocks, which type of rock are the following usually **made of**?

hardness


igneous rocks

★★★


sedimentary rocks

★★★


metamorphic rocks

★★☆



building materials

9. Different types of rocks have different colours, grain sizes and hardness. They have different uses in our daily lives.

Based on the characteristics of rocks, which type of rock are the following usually **made of**?

hardness



igneous rocks

★★★



sedimentary rocks

★★★



metamorphic rocks

★★☆



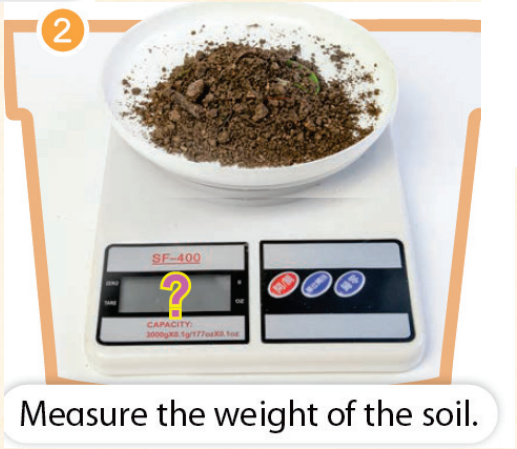
sculptures

10. Soil is formed when rocks break down. The components of it include soil particles, animal and plant matter, water and air.

Observe the soil using the following method.
What **components** does the soil contain?









10. Soil is formed when rocks break down. The components of it include soil particles, animal and plant matter, water and air.

Observe the soil using the following method.
What **components** does the soil include?



the components of soil include:

- soil particles of different sizes
- animal and plant matter
- water
- air

11. Soil is divided into sandy soil, loam and clay soil based on particle size.

What **types** of soil can be classified based on particle size?
What are the **characteristics** of each type?

	 sandy soil	 loam	 clay soil
particle size	largest / medium-sized / smallest	largest / medium-sized / smallest	largest / medium-sized / smallest
drainage capacity	highest / moderate / lowest	highest / moderate / lowest	highest / moderate / lowest
ability to retain nutrients	higher / moderate / lower	higher / moderate / lower	higher / moderate / lower

12. The drainage capacity of the soil affects the types of plants it is suitable for growing.

What kind of soil is best for growing cacti? What kind is best for growing roses?

	 sandy soil	 loam	 clay soil
particle size	largest / medium-sized / smallest	largest / medium-sized / smallest	largest / medium-sized / smallest
drainage capacity	highest / moderate / lowest	highest / moderate / lowest	highest / moderate / lowest
ability to retain nutrients	higher / moderate / lower	higher / moderate / lower	higher / moderate / lower

13. Rocks and soil are precious resources. We should cherish and protect them.

Why are these children **not** doing **right**?





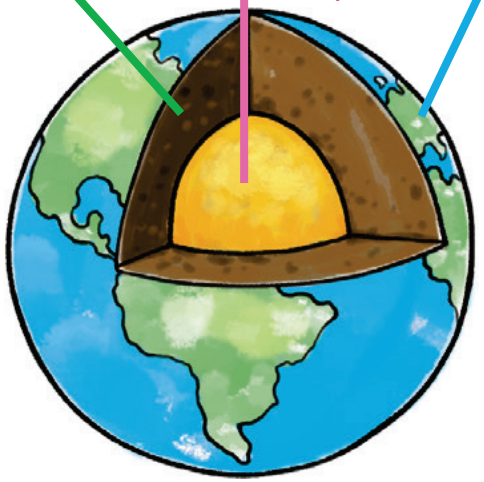
He takes away the ___rock___ when visiting the geopark.

She litters and ___pollutes___ the soil.

I have learned

Structure of the Earth

- **Mantle**
  - hot flowing magma
- **Core**
  - centre of the Earth, extremely high temperature
- **Crust**
  - outermost layer, divided into many plates



Rocks

- Igneous rocks
- Sedimentary rocks
- Metamorphic rocks

different colours, grain sizes and hardness

Soil

- Sandy soil
- Loam
- Clay soil

different drainage capacity